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**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 5

Data Sources & Data Collection

Learning outcomes

By the end of this topic you should be able to;

- *Identify sources of geographical data*
- *Know geographical data collectors and providers*



5.1. Introduction

Word processors, spreadsheets and graphics packages are programs that let you create and edit digital data. Each type of application saves its data into a particular file format. For example, a graphics program will let you save your drawing as a '.jpg' JPEG image, word processors let you save your document as an '.odt' OpenDocument or '.doc' Word Document, and so on. Just like these other applications, GIS Applications can store their data in files on the computer hard disk. There are a number of different file formats for GIS data, but the most common one is probably the 'shape file'. The name is a little odd in that although we call it a shape file (singular), it actually consists of at least three different files that work together to store your digital vector data. Many GIS Applications are also



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able to store digital data inside a database. In general, storing GIS data in a database is a good solution because the database can store large amounts of data efficiently and can provide data to the GIS Application quickly. Using a database also allows many people to work with the same vector data layers at the same time. Setting up a database to store GIS data is more complicated than using shapefiles.

Table 1: The basic files that together make up a 'shapefile'.



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Extension	Description
.shp	The geometry of vector features are stored in this file.
.dbf	The attributes of vector features are stored in this file.
.shx	This file is an index that helps the GIS Application to find features more quickly.





5.2. Sources of Data

Creating a GIS database is a daunting task. It involves data capture, verification, and structuring processes. Raw geographical data is available in many different analogue and digital forms such as maps, aerial photographs, satellite images, or tables.

Creation of a spatial database may involve the following:

- Acquire digital database from a supplier
- Digitize existing analogue data
- Carry out one's survey of geographical entities
- Interpolate from point observations to continuous surfaces

All data must be placed in a common coordinate system and well coded to be stored in the internal database of the GIS, resulting in a current, complete database for subsequent spa-



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tial analysis and modelling. As previously identified, two types of data are input into a GIS, spatial and attribute. The data input process is the operation of encoding both types of data into the GIS database formats. The creation of a clean digital database is the most important and time consuming task upon which the usefulness of the GIS depends. The establishment and maintenance of a robust spatial database is the cornerstone of a successful GIS implementation.

As well, the digital data is the most expensive part of the GIS. Yet often, not enough attention is given to the quality of the data or the processes by which they are prepared for automation. The general consensus among the GIS community is that 60 to 80 % of the cost incurred during implementation of GIS technology lies in data acquisition, data compilation and



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database development.

A wide variety of data sources exist for both spatial and attribute data. The most common general sources for spatial data are:

1. hard copy maps;
2. aerial photographs;
3. remotely-sensed imagery: derived from optical and digital remote sensing systems mounted on aircraft and satellite platforms provide much spatial information over many levels of spatial and temporal resolution
 - Stereo aerial photographs are overlapping analogue images that have many applications.
 - They are a major source of data for human interpretation and mapping of geology, soil, vegetation, or



land cover.

- They are also valuable background documents for placing other spatial data in a geographical context
- Digital photogrammetry and orthophoto mapping provide data on terrain elevation and land cover directly in digital form.
- Wide range of scanners on satellites provide digital data directly:

Passive systems: sense reflected and emitted radiation

Active systems: side-band radar scanners, laser altimeter scanners, sonar scanners point data samples from surveys;

4. existing digital data files.



5. Traditional survey methods of ground or field surveys aimed at recording sample values at known locations using instruments ranging from questionnaires , soil augers, to automatic chemical probes.

Existing hard copy maps, e.g. sometimes referred to as analogue maps, provide the most popular source for any GIS project.

Potential users should be aware that while there are many private sector firms specializing in providing digital data, federal, provincial and state government agencies are an excellent source of data. Because of the large costs associated with data capture and input, government departments are often the only agencies with financial resources and manpower funding to invest in data compilation. An inherent advantage of digital data from government agencies is its cost. It is typically inexpensive.



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However, this is often offset by the data's accuracy and quality. Attribute data has an even wider variety of data sources. Any textual or tabular data than can be referenced to a geographic feature, e.g. a point, line, or area, can be input into a GIS. Attribute data is usually input by manual keying or via a bulk loading utility of the DBMS software. ASCII format is a de facto standard for the transfer and conversion of attribute information.

It is important that all spatial data in a GIS is located with respect to a common frame. Placing these in a common system in GIS is attained through georeferencing. This frame is provided in one of a number of geodetic coordinate systems:

- Plane Orthogonal Cartesian Coordinates oriented conventionally north-south and east-west and derived from geo-



graphical coordinates

- Geographical (spherical) coordinates on ellipsoids for mapping the true curved surface of the earth

5.3. Geographic data collectors and providers

5.3.1. Systematic data collectors

- Government agencies: national and regional mapping agencies
 - Have the mandate to define and maintain the map spheroids, projections, grid origins, orientation and spacing, reference sea levels, data measurements and collection standards
 - Carry out cadastral surveys and registration of utilities in a systematic manner



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- Systematic collection of remote sensing data is carried out by government and private remote sensing corporations in some countries such as Canada, USA, Russia, India, France and Japan

5.3.2. Ad hoc Data Collectors

- Data collected by private surveyors, commercial organizations, market researchers, political organizations, or academic institutions
 - Data are for specific purposes
 - Access to these data is often restricted because of commercial interests



5.3.3. Data providers

- Offer and sell geographical data in a variety of detail, format, scales and structures.
- Sell data that they have processed but may not have acquired e.g. Regional Centre for Services in Surveying and Mapping in Kenya.
- Some data is provided free e.g. elevations, roads, towns, rivers.

5.4. Acquiring digital datasets

A user should consider the following characteristics of data to ensure they are compatible with the application:

- The currency of the data,



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- The length of the record,
- Scale of the data,
- Georeferencing system used
- Data collection technique and sampling strategy
- The quality of data collected
- Data classification and interpolation methods used
- Size and shape of the individual mapping units

After compatibility, the next issue is transfer of the data from the source to the GIS

- Encoding from source format to target format may be necessary
- Standards of interoperability have been developed , systems reading other systems' format



- Remote sensing data may need pre-processing before it can be input to a GIS i.e. Resolution adjustment, cartographic projection, classification, raster-vector conversion etc.

Data may also be gotten online with the Internet or Intranet

5.5. Creating digital datasets by manual input

Manual input of data into a GIS involves four (4) main stages:

1. **Entering the spatial data:** Geographical data in entity model are in terms of points, lines, or areas/pixels defined by a series of coordinates. Its simplest way is to type the coordinates into a file or input program of a GIS. This has been reduced by use of digitizers, scanners, or stereoplotters to encode the X and Y coordinates of desired points
2. **Entering the attribute data:** Attribute data are non-



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spatial characteristics of the spatial entity that need to be handled by GIS. Few attributes may be added at input of the spatial entity. Others come from different sources such as paper records, existing databases, spread sheets. They may be input manually, or imported into the GIS using a standard transfer, if in a relational database, an identifier is included in the attribute record to link the spatial and attribute data

3. **Spatial data and attribute verification and editing:**

It is important to check for errors, inaccuracies, omissions, etc. prior to linking the spatial and attribute data. The best way to check for errors is to produce a computer plot or print preferably on a translucent or thin paper and visually compare with the original print. Certain GIS show



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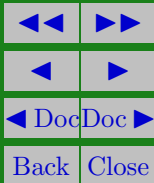
topological or identifier errors directly by colour coding the screen image. Attributes may be visually checked by printing out files or by eye; computer programs can be used to scan for gross errors. Errors that may arise during capture of spatial and attribute data may be grouped as follows:

- Spatial data are incomplete or double
- Spatial data in the wrong place
- Spatial data defined using too many coordinates
- Spatial data at the wrong scale: occurs during geo-referencing
- Spatial data are distorted: base maps not in correct scale

Typical errors arising in digitizing data are



- Dangling chain: (undershoot or overshoot)
 - Unlabelled polygon: due to duplicate digitizing of a single chain
 - Conflicting labels for a polygon: in case of multiple seed points for a given polygon, e.g. a group of islands belonging to a single polygon
 - Line segments with the same left and right labels; due to mislabelling of adjacent polygons
 - * The errors may be addressed through the editing and updating functions supported directly by most GIS.
 - * Data editing is usually undertaken by viewing the portion of the map containing errors on the screen and correcting them through





the software using the keyboard, screen cursor controlled by mouse, or a small digitizer tablet

- * Minor location errors can be corrected by moving the spatial entity through the screen cursor or by entering a new location on the digitizing tablet
- * Data scaling problems can be overcome by applying a simple numerical functions, e.g. through georeferencing
- * Excess coordinates can be removed through “weeding algorithms, with visual appearance being improved using splines.
- * Attribute values and spatial errors in raster



are edited by changing values of the faulty cell followed by inputting the correct attribute value.

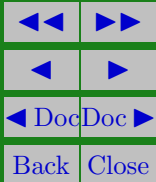
4. **Linking the spatial to the attribute data:** It is available in GIS with relational database structures and attained by use of identifiers which are common both in the spatial record and the attribute record. Identifiers are either automatically generated or added manually during digitizing or scanning/ vectorization process. Polygon identifiers are usually added after generation of topology. After the data capture process, we have geometrically correct raster data, or topologically and geometrically correct vector data. Most GIS accept both kinds of data. Structuring of the database as a whole is essential to optimize



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on storage requirement and performance of the GIS though sensible ordering and data compression techniques.





Revision Questions

EXERCISE 1.  Differentiate between passive and active systems.

Example . Identify errors in data files.

Solution: Revise. ☐

EXERCISE 2.  Describe the stages of creating digital datasets.



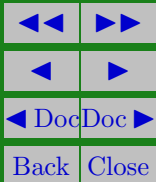
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Solutions to Exercises

Exercise 1. Revise.

Exercise 1





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Exercise 2. Revise.

Exercise 2

